Variable Scope in C

Visibility, Lifetime & Accessibility Rules

A Concise 8-Page Educational Resource

1. The Concept of Scope

Scope in C is a region of the program where a defined variable can be "seen" and used. It determines the **Visibility** and **Lifetime** of your data.

> **Key Benefit:** Scope helps prevent "Naming Conflicts" – you can have two variables named x as long as they are in different scopes.

In C, there are four primary types of scope: Local, Global, Block, and Function.

2. Local Scope

Variables declared **inside** a function or a block are called local variables. They are private to that function.

```
void myFunction() {
    int x = 10; // Local Scope
    printf("%d", x);
}
// printf("%d", x); // ERROR: x is invisible here
```

Characteristics:

- Created when the function starts.
- Destroyed when the function ends.
- Stored in the **Stack** memory.

3. Global Scope

Variables declared **outside** of all functions are global variables. They are accessible throughout the entire program.

```
int count = 5; // Global Level

void show() {
    printf("%d", count); // Accessible
}

int main() {
    show();
    return 0;
}
```

Lifetime: They exist from the start of the program until the program closes.

4. Block Scope

A variable declared inside a set of curly braces `{ }` is restricted to that specific block. This applies to `if`, `for`, and `while` loops.

```
if(1) {
    int blockVar = 50;
}
// blockVar cannot be accessed here
```

Note: In modern C (C99+), variables declared inside a `for` loop like `for(int i=0;...)` have block scope restricted to the loop.

5. Shadowing (Priority)

What happens if a Local and Global variable share the same name? In C, the **Local variable takes higher priority** inside its block.

```
int x = 100; // Global

int main() {
    int x = 10; // Local "Shadows" Global
    printf("%d", x); // Output: 10
    return 0;
}
```

This is called Name Shadowing or Variable Hiding.

6. Comparison Table

Scope Type	Location	Visibility
Local	Inside Function	Single Function only
Global	Outside Functions	Entire File/Program
Block	Inside { }	Specific Block only

Function Scope: Applies only to labels (used with `goto`) and function names themselves which are visible throughout the file.

Module Complete

Understanding scope is essential for writing safe code and managing memory efficiently in C programming.

*** End of 8-Page Scope Guide ***